

Parameter Bias in ETAS Models under Short-Term Incompleteness: A Bayesian Approach

MATH70088 Research Proposal

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1. Introduction and Research Questions

The Epidemic Type Aftershock Sequence (ETAS) model is a widely used point-process model for earthquake occurrence and aftershock triggering. In the temporal formulation considered here, the conditional intensity at time t is

$$\lambda(t \mid \mathcal{H}_t) = \mu + \sum_{t_i < t} \kappa(M_i) g(t - t_i),$$

where μ is the background seismicity rate, $\kappa(M_i) = K \cdot 10^{\alpha(M_i - M_{\text{cut}})}$ is the productivity function, and $g(\Delta t) = \frac{p-1}{c} \left(1 + \frac{\Delta t}{c}\right)^{-p}$ is a normalised Omori-type temporal kernel. The parameters K and α govern triggering productivity, while c and p control temporal decay.

A standard assumption in ETAS inference is that all earthquakes above a fixed magnitude threshold are completely observed. In practice, this assumption is often violated immediately after a large mainshock, when overlapping waveforms and network saturation reduce the detectability of smaller aftershocks. The effective completeness magnitude therefore becomes time-dependent, $M_c(t)$, before gradually returning to its baseline level. Ignoring this short-term incompleteness means that the observed catalogue captures only a partial record of early aftershock triggering, potentially inducing systematic bias in ETAS parameter estimates.

This project investigates the inferential consequences of short-term incompleteness for Bayesian ETAS models. Specifically, it examines whether a model that explicitly accounts for time-varying completeness improves parameter recovery and whether the resulting Bayesian credible intervals achieve appropriate frequentist coverage. This focus on uncertainty calibration, rather than forecasting performance, is motivated by applications such as seismic hazard assessment, where reliable quantification of parameter uncertainty is as important as accurate point estimation. Accordingly, this study addresses two research questions:

- RQ1.** To what extent does short-term aftershock incompleteness bias ETAS parameter estimates?
- RQ2.** Can modelling time-varying completeness within a Bayesian ETAS framework improve parameter recovery and uncertainty calibration?

2. Literature Review and Gap Analysis

The sensitivity of ETAS inference to catalogue completeness is well documented in the seismological literature [1, 2, 3, 4]. The effective completeness threshold $M_c(t)$ rises sharply immediately after large mainshocks, as overlapping waveforms and network saturation reduce the detectability of small aftershocks [3, 4]. For example, approximately 25% of events above $M \geq 2$ were missing from the Southern Californian catalogue following the 1992 Landers earthquake [3, 5]. These findings suggest that treating completeness as static can distort inference on ETAS productivity and temporal decay parameters. Seif et al. [4] quantified this effect through simulation, showing that ignoring short-term incompleteness leads to systematic underestimation of parameter uncertainty.

Several strategies have been proposed to address this. Raising the fixed cutoff M_{cut} improves observational reliability but discards the small early aftershocks that most strongly constrain short-term triggering behaviour [6, 7]. Moreover, parameter estimates can depend sensitively on the chosen threshold, introducing an additional source of bias [8, 9, 10]. Alternative approaches include excluding early post-mainshock time windows [11, 12] and constructing sequence-specific functional forms for $M_c(t)$ [13]. Mizrahi et al. [14] took a further step by embedding time-dependent completeness directly into the ETAS likelihood and demonstrated improved short-term forecasting accuracy. Collectively, these studies show that time-varying completeness can improve predictive performance, but they leave broader questions about parameter recovery and uncertainty calibration unresolved.

The inferential consequences of incompleteness for Bayesian ETAS models have begun to receive direct attention. Naylor et al. [15] showed, using synthetic catalogues within the `ETAS.inlabru` framework, that rate-dependent incompleteness has a significant and detrimental effect on ETAS posteriors, but explicitly noted that resolving this lay beyond the scope of that study. Kamranzad et al. [16] subsequently incorporated time-varying completeness into the ETAS likelihood within the same framework. Through synthetic experiments, they demonstrated that the modified model reduces parameter bias and recovers posterior distributions more accurately than the standard model, even when the productivity parameters K and α are difficult to distinguish.

Despite this progress, it remains unclear whether Bayesian credible intervals from an incompleteness-aware ETAS model achieve appropriate frequentist coverage. Kamranzad et al. [16] assess posterior recovery across individual simulation scenarios but do not report coverage probabilities across repeated simulations with known ground truth. This leaves an important inferential gap, because reduced posterior bias does not by itself guarantee well-calibrated uncertainty. This issue is especially relevant when $M_c(t)$ and the productivity parameter K are partially confounded. Poorly calibrated credible intervals would weaken the reliability of uncertainty quantification in operational earthquake forecasting and seismic hazard assessment.

3. Methodology

To address this coverage gap, the study adopts a three-stage design. The primary inferential conclusions are drawn from a controlled simulation study, while the empirical application serves as an illustrative case study.

3.1. Stage 1: Simulation-Based Bias Quantification

Synthetic ETAS catalogues will be generated under known parameter values $\theta = (\mu, K, c, p, \alpha)$ using Ogata’s thinning algorithm [17]. Short-term incompleteness will be introduced via a time-varying threshold $M_c(t)$ based on empirical recovery laws [13, 14, 18]. A static-threshold ETAS model will be fitted to both complete and incomplete catalogues generated from the same latent process, in order to distinguish bias induced by incompleteness from bias due to fixed-threshold misspecification. Performance will be evaluated across repeated simulations using parameter bias, RMSE, and the frequentist coverage of nominal 95% Bayesian credible intervals.

3.2. Stage 2: Bayesian Time-Varying ETAS Model

The model will be implemented in R using `inlabru`/R-INLA, treating the observed process as a thinned version of a latent ETAS process. A Bayesian framework is adopted because it produces full posterior distributions over parameters, enabling direct evaluation of credible-interval coverage rather than relying on asymptotic approximations. Following Mizrahi et al. [14] and Kamranzad et al. [16], the model introduces a time-varying detectable fraction $\Psi(t) = \min(1, 10^{-b_{GR}(M_c(t) - M_{cut})})$. The observed intensity is then written as $\lambda_{obs}(t) = \lambda_{true}(t) \Psi(t)$, allowing the likelihood to account for missing events without discarding data. A parametric form for $M_c(t)$ is adopted following Helmstetter et al. [18], in which the completeness threshold decays as a log-linear function of time since the mainshock. This reduces the number of free parameters and limits confounding with the productivity parameter K .

3.3. Stage 3: Empirical Application

The framework will be applied to the 2019 Ridgecrest sequence (M6.4 foreshock, M7.1 mainshock) using records from the SCECDC catalogue, which provides a well-documented case of short-term

incompleteness (Figure 1). Model adequacy will be assessed using transformed time residuals, posterior predictive checks, and WAIC.

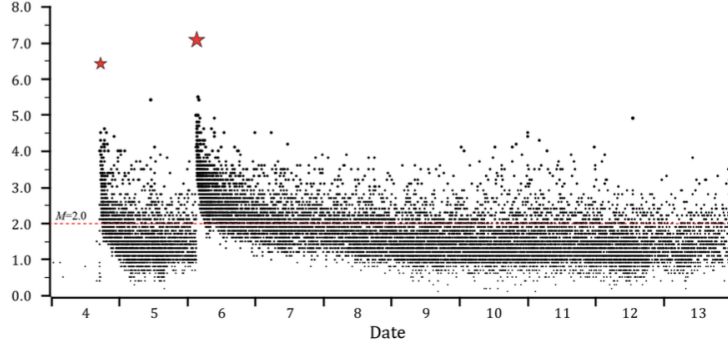


Figure 1: Magnitude–time plot for the first week of the Ridgecrest sequence, illustrating short-term aftershock incompleteness after the major events [19].

4. Challenges and Limitations

A key methodological challenge is parameter identifiability between the ETAS productivity parameters and the time-varying completeness function $M_c(t)$. To address this, the model will be evaluated first in simulation settings with known ground truth, and sensitivity analyses will be used to assess the robustness of inference under alternative prior and completeness specifications. Another limitation is that the assumed parametric form of $M_c(t)$ may not fully capture the true recovery of detectability after a mainshock. This will be assessed by considering alternative parametric specifications where feasible. Finally, the empirical application is limited to the 2019 Ridgecrest sequence, so the case study is used mainly as an illustration rather than as a basis for broad generalization across seismic settings.

5. Timeline and Deliverables

Weeks	Phase	Deliverables
1–3	Simulation design	Simulation framework finalised, ETAS parameter settings fixed, and incompleteness scenarios defined.
4–6	Model implementation	Baseline static-threshold and time-varying ETAS models implemented in R using <code>inlabru</code> /R-INLA.
7–8	Simulation evaluation	Comparative results summarised using bias, RMSE, and coverage across repeated simulations.
9–10	Empirical application	Ridgecrest case study completed, with fitted models and diagnostic results.
11	Sensitivity analysis	Robustness checks completed for priors, completeness settings, and model specification.
12	Final write-up	Final report, figures, and summary of main findings completed.

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